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Lens assembly

Abstract

A lens assembly includes a first lens, a second lens, a third lens, a fourth lens, and a fifth lens. The first lens is a meniscus lens with negative refractive power. The second lens is with positive refractive power and includes a convex surface facing an image side. The third lens is a meniscus lens with positive refractive power. The fourth lens is with positive refractive power. The fifth lens is with refractive power and includes a convex surface facing an object side. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from an object side to the image side along an optical axis.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The invention relates to a lens assembly.

Description of the Related Art

(2) The current development trend of a lens assembly is toward miniaturization and high resolution. Additionally, the lens assembly is developed to resist environmental temperature change in accordance with different application requirements. However, the known lens assembly can't satisfy such requirements. Therefore, the lens assembly needs a new structure in order to meet the requirements of miniaturization, high resolution, and resisted environmental temperature change at the same time.

BRIEF SUMMARY OF THE INVENTION

(3) The invention provides a lens assembly to solve the above problems. The lens assembly of the invention is provided with characteristics of a shortened total lens length, a decreased F-number, an increased resolution, a resisted environmental temperature change, and still has a good optical performance.

(4) The lens assembly in accordance with an exemplary embodiment of the invention includes a first lens, a second lens, a third lens, a fourth lens, and a fifth lens. The first lens is a meniscus lens with negative refractive power. The second lens is with positive refractive power and includes a convex surface facing an image side. The third lens is a meniscus lens with positive refractive power. The fourth lens is with positive refractive power. The fifth lens is with refractive power and includes a convex surface facing an object side. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from the object side to the image side along an optical axis.

(5) In another exemplary embodiment, the first lens includes a convex surface facing the object side and a concave surface facing the image side, the second lens further includes a concave surface facing the object side, and the third lens includes a concave surface facing the object side and a convex surface facing the image side.

(6) In yet another exemplary embodiment, the fourth lens is a meniscus lens and includes a convex

surface facing the object side and a concave surface facing the image side and the fifth lens is a biconvex lens with positive refractive power and further includes a convex surface facing the image side.

(7) In another exemplary embodiment, the lens assembly further includes a stop disposed between the second lens and the third lens.

(8) In yet another exemplary embodiment, the lens assembly satisfies: $0.1 < \Phi_{ST}/TTL < 0.5$; wherein Φ_{ST} is an aperture diameter of the stop and TTL is an interval from an object side surface of the first lens to an image plane along the optical axis.

(9) In another exemplary embodiment, the fifth lens is a single lens and an air gap is disposed between the fourth lens and the fifth lens.

(10) In yet another exemplary embodiment, the lens assembly satisfies: $-3 < f_1/f < -1.5$; $0.5 < ET_5/T_5 < 1.5$; $0.3 < \Phi_2/\Phi_3 < 0.8$; wherein f_1 is an effective focal length of the first lens, f is an effective focal length of the lens assembly, ET_5 is a periphery thickness of the fifth lens, T_5 is a thickness of the fifth lens along the optical axis, Φ_2 is a maximum effective optical diameter of the second lens, and Φ_3 is a maximum effective optical diameter of the third lens.

(11) In another exemplary embodiment, the lens assembly further includes a sixth lens and a seventh lens, wherein the sixth lens is a meniscus lens with negative refractive power and disposed between the first lens and the second lens, the seventh lens is a meniscus lens and disposed between the third lens and the fourth lens, and the fifth lens is a meniscus lens with positive refractive power and further includes a concave surface facing the image side.

(12) In yet another exemplary embodiment, the first lens includes a convex surface facing the object side and a concave surface facing the image side, the second lens further includes another convex surface facing the object side, the third lens includes a concave surface facing the object side and a convex surface facing the image side, and the fourth lens includes a convex surface facing the image side.

(13) In another exemplary embodiment, the sixth lens includes a concave surface facing the object side and a convex surface facing the image side, and the seventh lens is with positive refractive power and includes a convex surface facing the object side and a concave surface facing the image side.

(14) In yet another exemplary embodiment, the fourth lens further includes another convex surface facing the object side.

(15) In another exemplary embodiment, the fourth lens further includes a concave surface facing the object side.

(16) In yet another exemplary embodiment, the sixth lens includes a convex surface facing the object side and a concave surface facing the image side, and the seventh lens is with negative refractive power and includes a concave surface facing the object side and a convex surface facing the image side.

(17) In another exemplary embodiment, the fourth lens further includes a concave surface facing the object side.

(18) In yet another exemplary embodiment, the lens assembly satisfies at least one of the following conditions: $65 \text{ mm} \cdot \sup{.2^\circ} \leq ((f \times FOV \times \pi)/360) \times TTL \leq 70 \text{ mm} \cdot \sup{.2^\circ}$; $L1D/R12 \leq 1.8$; $0.4 \leq L6T1/L6T2 \leq 0.8$; $1.54 \leq Nd6 \leq 1.68$; $1.54 \leq Nd7 \leq 1.68$; $9 \text{ mm} \leq Nd2 \times f2 \leq 18 \text{ mm}$; wherein f is an effective focal length of the lens assembly, FOV is a field of view of the lens assembly, TTL is an interval from an object side surface of the first lens to an image plane along the optical axis, $L1D$ is an effective optical diameter of an image side surface of the first lens, $R12$ is a radius of curvature of the image side surface of the first lens, $L6T1$ is a thickness of the sixth lens along the optical axis, $L6T2$ is a periphery thickness of the sixth lens, $Nd6$ is an index of refraction of the sixth lens, $Nd7$ is an index of refraction of the seventh lens, $Nd2$ is an index of refraction of the second lens, and $f2$ is an effective focal length of the second lens.

(19) In another exemplary embodiment, the lens assembly further includes a stop disposed between

the sixth lens and the third lens, wherein the lens assembly satisfies at least one of the following conditions: $25\text{ mm} \leq \text{Nd3} \times \text{f3} \leq 55\text{ mm}$; $20\text{ mm} \leq \text{NdLG} \times \text{fLG} \leq 40\text{ mm}$; $13\text{ mm} \leq \text{fAS2} \leq 40\text{ mm}$; $3.5\text{ mm} \leq \text{fAS} \leq 5.5\text{ mm}$; wherein Nd3 is an index of refraction of the third lens, f3 is an effective focal length of the third lens, NdLG is an index of refraction of the lens closest to the image side, fLG is an effective focal length of the lens closest to the image side, fAS2 is an effective focal length of the lens second away from the stop to the image side, and fAS is an effective focal length of a combination of all lenses between the stop and the image side.

(20) The lens assembly in accordance with another exemplary embodiment of the invention includes a first lens, a second lens, a third lens, a fourth lens, and a fifth lens. The first lens is a meniscus lens with negative refractive power. The second lens is with positive refractive power and includes a convex surface facing an image side. The third lens is a meniscus lens with positive refractive power. The fourth lens is with positive refractive power. The fifth lens is with negative refractive power. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from an object side to the image side along an optical axis.

(21) In another exemplary embodiment, the lens assembly further includes a stop disposed between the second lens and the third lens, wherein the first lens includes a convex surface facing the object side and a concave surface facing the image side, the second lens is a meniscus lens and further includes a concave surface facing the object side, the third lens includes a concave surface facing the object side and a convex surface facing the image side, the fourth lens is a biconvex lens and includes a convex surface facing the object side and another convex surface facing the image side, and the fifth lens is a meniscus lens and includes a concave surface facing the object side and a convex surface facing the image side.

(22) In yet another exemplary embodiment, the lens assembly satisfies at least one of following conditions: $0.1 < \Phi_{\text{ST}}/\text{TTL} < 0.5$; $-3 < \text{f1}/\text{f} < -1.5$; $0.5 < \text{ET5}/\text{T5} < 1.5$; $0.3 < \Phi_2/\Phi_3 < 0.8$; wherein Φ_{ST} is an aperture diameter of the stop, TTL is an interval from an object side surface of the first lens to an image plane along the optical axis, f1 is an effective focal length of the first lens, f is an effective focal length of the lens assembly, ET5 is a periphery thickness of the fifth lens, T5 is a thickness of the fifth lens along the optical axis, Φ_2 is a maximum effective optical diameter of the second lens, and Φ_3 is a maximum effective optical diameter of the third lens.

(23) A detailed description is given in the following embodiments with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

(2) FIG. 1 is a lens layout diagram of a lens assembly in accordance with a first embodiment of the invention;

(3) FIG. 2A, FIG. 2B, and FIG. 2C depict a longitudinal aberration diagram, a field curvature diagram, and a distortion diagram of the lens assembly in accordance with the first embodiment of the invention, respectively;

(4) FIG. 3 is a lens layout diagram of a lens assembly in accordance with a second embodiment of the invention;

(5) FIG. 4A, FIG. 4B, and FIG. 4C depict a longitudinal aberration diagram, a field curvature diagram, and a distortion diagram of the lens assembly in accordance with the second embodiment of the invention, respectively;

(6) FIG. 5 is a lens layout diagram of a lens assembly in accordance with a third embodiment of the invention;

- (7) FIG. 6A, FIG. 6B, and FIG. 6C depict a longitudinal aberration diagram, a field curvature diagram, and a distortion diagram of the lens assembly in accordance with the third embodiment of the invention, respectively;
- (8) FIG. 7 is a lens layout diagram of a lens assembly in accordance with a fourth embodiment of the invention;
- (9) FIG. 8A, FIG. 8B, FIG. 8C, and FIG. 8D depict a longitudinal aberration diagram, a field curvature diagram, a distortion diagram, and a lateral color diagram of the lens assembly in accordance with the fourth embodiment of the invention, respectively;
- (10) FIG. 9 is a lens layout diagram of a lens assembly in accordance with a fifth embodiment of the invention;
- (11) FIG. 10A, FIG. 10B, FIG. 10C, and FIG. 10D depict a longitudinal aberration diagram, a field curvature diagram, a distortion diagram, and a lateral color diagram of the lens assembly in accordance with the fifth embodiment of the invention, respectively;
- (12) FIG. 11 is a lens layout diagram of a lens assembly in accordance with a sixth embodiment of the invention; and
- (13) FIG. 12A, FIG. 12B, FIG. 12C, and FIG. 12D depict a longitudinal aberration diagram, a field curvature diagram, a distortion diagram, and a lateral color diagram of the lens assembly in accordance with the sixth embodiment of the invention, respectively.

DETAILED DESCRIPTION OF THE INVENTION

- (14) The following description is made for the purpose of illustrating the general principles of the invention and should not be taken in a limiting sense. The scope of the invention is best determined by reference to the appended claims.
- (15) The present invention provides a lens assembly including a first lens, a second lens, a third lens, a fourth lens, and a fifth lens. The first lens is a meniscus lens with negative refractive power. The second lens is with positive refractive power and includes a convex surface facing an image side. The third lens is a meniscus lens with positive refractive power. The fourth lens is with positive refractive power. The fifth lens is with refractive power and includes a convex surface facing an object side. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from the object side to the image side along an optical axis.
- (16) The present invention provides another lens assembly including a first lens, a second lens, a third lens, a fourth lens, and a fifth lens. The first lens is a meniscus lens with negative refractive power. The second lens is with positive refractive power and includes a convex surface facing an image side. The third lens is a meniscus lens with positive refractive power. The fourth lens is with positive refractive power. The fifth lens is with negative refractive power. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from an object side to the image side along an optical axis.
- (17) Referring to Table 1, Table 2, Table 4, Table 5, Table 7, and Table 8, wherein Table 1, Table 4, and Table 7 show optical specification in accordance with a first, second, and third embodiments of the invention, respectively and Table 2, Table 5, and Table 8 show aspheric coefficients of each aspheric lens in Table 1, Table 4, and Table 7, respectively.
- (18) FIG. 1, FIG. 3, and FIG. 5 are lens layout diagrams of the lens assemblies in accordance with the first, second, and third embodiments of the invention, respectively.
- (19) The first lenses L11, L21, L31 are meniscus lenses with negative refractive power and made of glass material, wherein the object side surfaces S11, S21, S31 are convex surfaces, the image side surfaces S12, S22, S32 are concave surfaces, and both of the object side surfaces S11, S21, S31 and image side surfaces S12, S22, S32 are spherical surfaces.
- (20) The second lenses L12, L22, L32 are meniscus lenses with positive refractive power and made of glass material, wherein the object side surfaces S13, S23, S33 are concave surfaces, the image side surfaces S14, S24, S34 are convex surfaces, and both of the object side surfaces S13, S23, S33 and image side surfaces S14, S24, S34 are aspheric surfaces.

- (21) The third lenses **L13**, **L23**, **L33** are meniscus lenses with positive refractive power and made of glass material, wherein the object side surfaces **S16**, **S26**, **S36** are concave surfaces, the image side surfaces **S17**, **S27**, **S37** are convex surfaces, and both of the object side surfaces **S16**, **S26**, **S36** and image side surfaces **S17**, **S27**, **S37** are spherical surfaces.
- (22) The fourth lenses **L14**, **L24**, **L34** are with positive refractive power and made of glass material, wherein the object side surfaces **S18**, **S28**, **S38** are convex surfaces and both of the object side surfaces **S18**, **S28**, **S38** and image side surfaces **S19**, **S29**, **S39** are spherical surfaces.
- (23) The fifth lenses **L15**, **L25**, **L35** are with positive refractive power and made of glass material, wherein the image side surfaces **S111**, **S211**, **S311** are convex surfaces and both of the object side surfaces **S110**, **S210**, **S310** and image side surfaces **S111**, **S211**, **S311** are aspheric surfaces.
- (24) In addition, the lens assemblies **1**, **2**, **3** satisfy at least one of the following conditions:
- $-3 < f1/f < -1.5$; (1)
 - $0.5 < ET5/T5 < 1.5$; (2)
 - $0.3 < \Phi2/\Phi3 < 0.8$; (3)
 - $0.1 < \Phi_{ST}/TTL < 0.5$; (4)
 - $L1D/R12 \leq 1.8$; (5)
 - $3.5 \text{ mm} \leq f_{AS} \leq 5.5 \text{ mm}$; (6)
- (25) wherein $f1$ is an effective focal length of the first lenses **L11**, **L21**, **L31** for the first to third embodiments, f is an effective focal length of the lens assemblies **1**, **2**, **3** for the first to third embodiments, $ET5$ is a periphery thickness of the fifth lenses **L15**, **L25**, **L35** for the first to third embodiments, $T5$ is a thickness of the fifth lenses **L15**, **L25**, **L35** along the optical axes **OA1**, **OA2**, **OA3** for the first to third embodiments, $\Phi2$ is a maximum effective optical diameter of the second lenses **L12**, **L22**, **L32** for the first to third embodiments, $\Phi3$ is a maximum effective optical diameter of the third lenses **L13**, **L23**, **L33** for the first to third embodiments, Φ_{ST} is an aperture diameter of the stops **ST1**, **ST2**, **ST3** for the first to third embodiments, TTL is an interval from object side surfaces **S11**, **S21**, **S31** of the first lenses **L11**, **L21**, **L31** to an image planes **IMA1**, **IMA2**, **IMA3** along the optical axes **OA1**, **OA2**, **OA3** for the first to third embodiments, $L1D$ is an effective optical diameter of the image side surfaces **S12**, **S22**, **S32** of the first lenses **L11**, **L21**, **L31** for the first to third embodiments, $R12$ is a radius of curvature of the image side surfaces **S12**, **S22**, **S32** of the first lenses **L11**, **L21**, **L31** for the first to third embodiments, and f_{AS} is an effective focal length of a combination of all lenses between the stops **ST1**, **ST2**, **ST3** and the image side for the first to third embodiments. With the lens assemblies **1**, **2**, **3** satisfying at least one of the above conditions (1)-(6), the total lens length can be effectively shortened, the resolution can be effectively increased, the environmental temperature change can be effectively resisted, and the aberration can be effectively corrected.
- (26) When the condition (1): $-3 < f1/f < -1.5$ is satisfied, the field of view can be increased effectively.
- (27) When the condition (2): $0.5 < ET5/T5 < 1.5$ is satisfied, the processability can be enhanced effectively.
- (28) When the condition (3): $0.3 < \Phi2/\Phi3 < 0.8$ is satisfied, the imaging range can be increased effectively.
- (29) When the condition (4): $0.1 < \Phi_{ST}/TTL < 0.5$ is satisfied, the F-number can be decreased effectively.
- (30) When the condition (5): $L1D/R12 \leq 1.8$ is satisfied, the processing cost of lenses can be decreased effectively.
- (31) When the condition (6): $3.5 \text{ mm} \leq f_{AS} \leq 5.5 \text{ mm}$ is satisfied, the sensitivity of the lens combination on both sides of the stop can be balanced effectively.
- (32) All glass lens design can effectively resist environmental temperature changes, so that the lens assembly still has good optical performance under high or low temperature environment, and is not easy to deteriorate after long term use and still has good optical performance.

(33) A detailed description of a lens assembly in accordance with a first embodiment of the invention is as follows. Referring to FIG. 1, the lens assembly 1 includes a first lens L11, a second lens L12, a stop ST1, a third lens L13, a fourth lens L14, a fifth lens L15, an optical filter OF1, and a cover glass CG1, all of which are arranged in order from an object side to an image side along an optical axis OA1. In operation, the light from the object side is imaged on an image plane IMA1.

(34) According to subject matter previously described herein, wherein: the fourth lens L14 is a meniscus lens, wherein the image side surface S19 is a concave surface; the fifth lens L15 is a biconvex lens, wherein the object side surface S110 is a convex surface; both of the object side surface S112 and image side surface S113 of the optical filter OF1 are plane surfaces; and both of the object side surface S114 and image side surface S115 of the cover glass CG1 are plane surfaces.

(35) With the above design of the lenses, stop ST1, and at least one of the conditions (1)-(6) satisfied, the lens assembly 1 can have an effective shortened total lens length, an effective increased resolution, an effective resisted environmental temperature change, and an effective corrected aberration.

(36) Table 1 shows the optical specification of the lens assembly 1 in FIG. 1.

(37) TABLE-US-00001

TABLE 1 Effective Focal Length = 2.858 mm F-number = 1.33 Total Lens Length = 15.405 mm Field of View = 141.24 degrees Radius of Effective Surface Curvature											
Thickness	Focal Length	Number (mm)	(mm)	Nd	Vd (mm)	Remark	S11	8.860	0.409	1.61	56.7
-5.998	L11	S12	2.505	2.213	S13	-4.551	1.730	1.82	42.5	11.200	L12
S14	-3.523	-0.512	S15	∞	2.006	ST1	S16	-8.841	2.054	1.75	51
9.945	L13	S17	-4.386	0.536	S18	5.788	1.589	1.69	54.6	25.433	L14
S19	7.736	1.161	S110	19.748	1.060	1.86	39.9	11.741	L15	S111	-19.208
1.782	S112	∞	0.300	1.52	64.2	OF1	S113	∞	0.500	S114	∞
0.500	1.52	64.2	CG1	S115	∞	0.076					

(38) The aspheric surface sag z of each aspheric lens in table 1 can be calculated by the following formula:

$$z = ch.\text{sup.}2 / \{1 + [1 - (k+1)c.\text{sup.}2h.\text{sup.}2].\text{sup.}1/2\} + Ah.\text{sup.}4 + Bh.\text{sup.}6 + Ch.\text{sup.}8$$

where c is curvature, h is the vertical distance from the lens surface to the optical axis, k is conic constant and A, B, and C are aspheric coefficients.

(39) In the first embodiment, the conic constant k and the aspheric coefficients A, B, C of each aspheric lens are shown in Table 2.

(40) TABLE-US-00002

TABLE 2 Surface Number k A B C S13											
0.00E+00	-6.981E-03	-4.562E-04	4.379E-05	S14	0.00E+00	3.203E-04	-1.811E-04	7.264E-05	S110	0.00E+00	-2.308E-03
1.187E-04	-2.379E-06	S111	0.00E+00	-1.032E-03	1.541E-04	-4.077E-06					

(41) Table 3 shows the parameters and condition values for conditions (1)-(6) in accordance with the first embodiment of the invention. It can be seen from Table 3 that the lens assembly 1 of the first embodiment satisfies the conditions (1)-(6).

(42) TABLE-US-00003

TABLE 3 ET5 0.589 mm T5 1.060 mm Φ2 3.749 mm Φ3 7.017 mm ΦST 3.742 mm L1D 4.30 mm fAS 4.943 mm f1/f -2.099 ET5/T5 0.556 Φ2/Φ3 0.534 ΦST/TTL 0.243 L1D/R12 1.72											

(43) In addition, the lens assembly 1 of the first embodiment can meet the requirements of optical performance as seen in FIGS. 2A-2C. It can be seen from FIG. 2A that the longitudinal aberration in the lens assembly 1 of the first embodiment ranges from -0.001 mm to 0.015 mm. It can be seen from FIG. 2B that the field curvature of tangential direction and sagittal direction in the lens assembly 1 of the first embodiment ranges from -0.04 mm to 0.05 mm. It can be seen from FIG. 2C that the distortion in the lens assembly 1 of the first embodiment ranges from -70% to 0%. It is obvious that the longitudinal aberration, the field curvature, and the distortion of the lens assembly 1 of the first embodiment can be corrected effectively. Therefore, the lens assembly 1 of the first embodiment is capable of good optical performance.

(44) Referring to FIG. 3, the lens assembly 2 includes a first lens L21, a second lens L22, a stop ST2, a third lens L23, a fourth lens L24, a fifth lens L25, an optical filter OF2, and a cover glass CG2, all of which are arranged in order from an object side to an image side along an optical axis

OA2. In operation, the light from the object side is imaged on an image plane IMA2.

(45) According to subject matter previously described herein, wherein: the fourth lens L24 is a meniscus lens, wherein the image side surface S29 is a concave surface; the fifth lens L25 is a biconvex lens, wherein the object side surface S210 is a convex surface; both of the object side surface S212 and image side surface S213 of the optical filter OF2 are plane surfaces; and both of the object side surface S214 and image side surface S215 of the cover glass CG2 are plane surfaces.

(46) With the above design of the lenses, stop ST2, and at least one of the conditions (1)-(6) satisfied, the lens assembly 2 can have an effective shortened total lens length, an effective increased resolution, an effective resisted environmental temperature change, and an effective corrected aberration.

(47) Table 4 shows the optical specification of the lens assembly 2 in FIG. 3.

(48) TABLE-US-00004 TABLE 4 Effective Focal Length = 2.829 mm F-number = 1.34 Total Lens Length = 15.192 mm Field of View = 148.00 degrees Effective Radius of Focal Surface Curvature Thickness Length Number (mm) (mm) Nd Vd (mm) Remark S21 8.113 0.448 1.61 56.7 -5.339 L21 S22 2.241 2.224 S23 -4.429 1.472 1.82 42.5 10.894 L22 S24 -3.366 -0.580 S25 ∞ 2.169 ST2 S26 -10.842 1.827 1.69 53.3 10.108 L23 S27 -4.497 0.105 S28 6.371 1.352 1.69 54.9 30.167 L24 S29 8.456 0.821 S210 12.705 2.106 1.86 39.9 11.247 L25 S211 -34.017 1.873 S212 ∞ 0.300 1.52 64.2 OF2 S213 ∞ 0.500 S214 ∞ 0.500 1.52 64.2 CG2 S215 ∞ 0.076

(49) The definition of aspheric surface sag z of each aspheric lens in Table 4 is the same as that of in Table 1, and is not described here again.

(50) In the second embodiment, the conic constant k and the aspheric coefficients A, B, C of each aspheric lens are shown in Table 5.

(51) TABLE-US-00005 TABLE 5 Surface Number k A B C S23 0.00E+00 -8.17E-03 -1.31E-03 -7.49E-05 S24 0.00E+00 -1.85E-03 -1.97E-04 0.00E+00 S210 0.00E+00 -4.54E-04 -1.57E-05 0.00E+00 S211 0.00E+00 9.83E-04 -3.33E-05 0.00E+00

(52) Table 6 shows the parameters and condition values for conditions (1)-(6) in accordance with the second embodiment of the invention. It can be seen from Table 6 that the lens assembly 2 of the second embodiment satisfies the conditions (1)-(6).

(53) TABLE-US-00006 TABLE 6 ET5 1.529 mm T5 2.106 mm Φ 2 3.742 mm Φ 3 6.998 mm Φ ST 3.742 mm L1D 4.04 mm fAS 4.876 mm f1/f -1.887 ET5/T5 0.726 Φ 2/ Φ 3 0.535 Φ ST/TTL 0.246 L1D/R12 1.80

(54) In addition, the lens assembly 2 of the second embodiment can meet the requirements of optical performance as seen in FIGS. 4A-4C. It can be seen from FIG. 4A that the longitudinal aberration in the lens assembly 2 of the second embodiment ranges from -0.01 mm to 0.04 mm. It can be seen from FIG. 4B that the field curvature of tangential direction and sagittal direction in the lens assembly 2 of the second embodiment ranges from -0.04 mm to 0.05 mm. It can be seen from FIG. 4C that the distortion in the lens assembly 2 of the second embodiment ranges from -70% to 0%. It is obvious that the longitudinal aberration, the field curvature, and the distortion of the lens assembly 2 of the second embodiment can be corrected effectively. Therefore, the lens assembly 2 of the second embodiment is capable of good optical performance.

(55) Referring to FIG. 5, the lens assembly 3 includes a first lens L31, a second lens L32, a stop ST3, a third lens L33, a fourth lens L34, a fifth lens L35, an optical filter OF3, and a cover glass CG3, all of which are arranged in order from an object side to an image side along an optical axis OA3. In operation, the light from the object side is imaged on an image plane IMA3.

(56) According to subject matter previously described herein, wherein: the fourth lens L34 is a biconvex lens, wherein the image side surface S39 is a convex surface; the fifth lens L35 is a meniscus lens, wherein the object side surface S310 is a concave surface; both of the object side surface S312 and image side surface S313 of the optical filter OF3 are plane surfaces; and both of the object side surface S314 and image side surface S315 of the cover glass CG3 are plane

surfaces.

(57) With the above design of the lenses, stop **ST3**, and at least one of the conditions (1)-(6) satisfied, the lens assembly **3** can have an effective shortened total lens length, an effective increased resolution, an effective resisted environmental temperature change, and an effective corrected aberration.

(58) Table 7 shows the optical specification of the lens assembly **3** in FIG. 5.

(59) TABLE-US-00007 TABLE 7 Effective Focal Length = 2.874 mm F-number = 1.33 Total Lens Length = 15.081 mm Field of View = 148.00 degrees Effective Radius of Focal Surface Curvature Thickness Length Number (mm) (mm) Nd Vd (mm) Remark S31 8.489 0.782 1.61 56.7 -5.026 L31 S32 2.140 2.046 S33 -4.793 1.452 1.82 42.5 10.422 L32 S34 -3.448 -0.569 S35 ∞ 2.203 ST3 S36 -10.206 1.667 1.69 53.2 10.588 L33 S37 -4.503 0.104 S38 9.396 2.380 1.69 53.2 7.762 L34 S39 -10.833 0.890 S310 -30.535 1.089 1.92 24 -49.981 L35 S311 -99.276 1.659 S312 ∞ 0.300 1.52 64.2 OF3 S313 ∞ 0.500 S314 ∞ 0.500 1.52 64.2 CG3 S315 ∞ 0.076

(60) The definition of aspheric surface sag z of each aspheric lens in Table 7 is the same as that of in Table 1, and is not described here again.

(61) In the third embodiment, the conic constant k and the aspheric coefficients A , B , C of each aspheric lens are shown in Table 8.

(62) TABLE-US-00008 TABLE 8 Surface Number k A B C S33 0.00E+00 -7.76E-03 -1.49E-03 0.00E+00 S34 0.00E+00 -1.83E-03 -1.85E-04 0.00E+00 S310 0.00E+00 -3.64E-03 0.00E+00 0.00E+00 S311 0.00E+00 -2.56E-03 0.00E+00 0.00E+00

(63) Table 9 shows the parameters and condition values for conditions (1)-(6) in accordance with the third embodiment of the invention. It can be seen from Table 9 that the lens assembly **3** of the third embodiment satisfies the conditions (1)-(6).

(64) TABLE-US-00009 TABLE 9 ET5 1.268 mm T5 1.089 mm Φ 2 3.755 mm Φ 3 6.721 mm Φ ST 3.759 mm L1D 3.78 mm fAS 4.793 mm f1/f -1.749 ET5/T5 1.164 Φ 2/ Φ 3 0.559 Φ ST/TTL 0.249 L1D/R12 1.77

(65) In addition, the lens assembly **3** of the third embodiment can meet the requirements of optical performance as seen in FIGS. 6A-6C. It can be seen from FIG. 6A that the longitudinal aberration in the lens assembly **3** of the third embodiment ranges from -0.001 mm to 0.045 mm. It can be seen from FIG. 6B that the field curvature of tangential direction and sagittal direction in the lens assembly **3** of the third embodiment ranges from -0.03 mm to 0.07 mm. It can be seen from FIG. 6C that the distortion in the lens assembly **3** of the third embodiment ranges from -70% to 0%. It is obvious that the longitudinal aberration, the field curvature, and the distortion of the lens assembly **3** of the third embodiment can be corrected effectively. Therefore, the lens assembly **3** of the third embodiment is capable of good optical performance.

(66) Referring to Table 10, Table 11, Table 13, Table 14, Table 16, and Table 17, wherein Table 10, Table 13, and Table 16 show optical specification in accordance with a fourth, fifth, and sixth embodiments of the invention, respectively and Table 11, Table 14, and Table 17 show aspheric coefficients of each aspheric lens in Table 10, Table 13, and Table 16, respectively.

(67) FIG. 7, FIG. 9, and FIG. 11 are lens layout diagrams of the lens assemblies in accordance with the fourth, fifth, and sixth embodiments of the invention, respectively.

(68) The first lenses **L41**, **L51**, **L61** are meniscus lenses with negative refractive power and made of glass material, wherein the object side surfaces **S41**, **S51**, **S61** are convex surfaces, the image side surfaces **S42**, **S52**, **S62** are concave surfaces, and both of the object side surfaces **S41**, **S51**, **S61** and image side surfaces **S42**, **S52**, **S62** are spherical surfaces.

(69) The sixth lenses **L46**, **L56**, **L66** are meniscus lenses with negative refractive power and made of plastic material, wherein both of the object side surfaces **S43**, **S53**, **S63** and image side surfaces **S44**, **S54**, **S64** are aspheric surfaces.

(70) The second lenses **L42**, **L52**, **L62** are biconvex lenses with positive refractive power and made of glass material, wherein the object side surfaces **S45**, **S55**, **S66** are convex surfaces, the image

side surfaces **S46**, **S56**, **S67** are convex surfaces, and both of the object side surfaces **S45**, **S55**, **S66** and image side surfaces **S46**, **S56**, **S67** are spherical surfaces.

(71) The third lenses **L43**, **L53**, **L63** are meniscus lens with positive refractive power, wherein the object side surfaces **S48**, **S58**, **S68** are concave surfaces which can reduce the light blocked by the optical filters **OF4**, **OF5**, **OF6** reflecting to the object side surfaces **S48**, **S58**, **S68** of the third lens **L43**, **L53**, **L63** and then reflected back to produce ghost image, and the image side surfaces **S49**, **S59**, **S69** are convex surfaces.

(72) The seventh lenses **L47**, **L57**, **L67** are meniscus lenses and made of plastic material, wherein both of the object side surfaces **S410**, **S510**, **S610** and image side surfaces **S411**, **S511**, **S611** are aspheric surfaces.

(73) The fourth lenses **L44**, **L54**, **L64** are with positive refractive power and can disperse lens sensitivity, wherein the image side surfaces **S413**, **S513**, **S613** are convex surfaces.

(74) The fifth lenses **L45**, **L55**, **L65** are meniscus lenses with positive refractive power and made of glass material, wherein the object side surfaces **S414**, **S514**, **S614** are convex surfaces, the image side surfaces **S415**, **S515**, **S615** are concave surfaces, both of which can reduce the light blocked by the optical filters **OF4**, **OF5**, **OF6** reflecting to the image side surfaces **S415**, **S515**, **S615** of the fifth lenses **L45**, **L55**, **L65** and then reflected back to produce ghost image focusing on the light source so that the ghost image overlaps the light source and is difficult to find, and both of the object side surfaces **S414**, **S514**, **S614** and image side surfaces **S415**, **S515**, **S615** are spherical surfaces.

(75) The ghost image can be effectively reduced when the object side surface of the third lens is a concave surface, the fourth lens has positive refractive power, and the image side surface of the fifth lens is a concave surface.

(76) In addition, the lens assemblies **4**, **5**, **6**, satisfy at least one of the conditions (2), (4), and the following conditions (7)-(14):

$$65 \text{ mm} \cdot \sup.2 \times \text{degrees} \leq ((f \times FOV \times \pi) / 360) \times TTL \leq 70 \text{ mm} \cdot \sup.2 \times \text{degrees}; \quad (7)$$

$$0.4 \leq L6T1 / L6T2 \leq 0.8; \quad (8)$$

$$1.54 \leq Nd6 \leq 1.68; \quad (9)$$

$$1.54 \leq Nd7 \leq 1.68; \quad (10)$$

$$9 \text{ mm} \leq Nd2 \times f2 \leq 18 \text{ mm}; \quad (11)$$

$$25 \text{ mm} \leq Nd3 \times f3 \leq 55 \text{ mm}; \quad (12)$$

$$20 \text{ mm} \leq NdLG \times fLG \leq 40 \text{ mm}; \quad (13)$$

$$13 \text{ mm} \leq fAS2 \leq 40 \text{ mm}; \quad (14)$$

(77) wherein f is an effective focal length of the lens assemblies **4**, **5**, **6** for the fourth to sixth embodiments, $f2$ is an effective focal length of the second lenses **L42**, **L52**, **L62** for the fourth to sixth embodiments, $f3$ is an effective focal length of the third lenses **L43**, **L53**, **L63** for the fourth to sixth embodiments, fLG is an effective focal length of the lenses **L45**, **L55**, **L65** closest to the image side for the fourth to sixth embodiments, FOV is a field of view of the lens assemblies **4**, **5**, **6** for the fourth to sixth embodiments, TTL is an interval from the object side surfaces **S41**, **S51**, **S61** of the first lenses **L41**, **L51**, **L61** to image planes **IMA4**, **IMA5**, **IMA6** along the optical axes **OA4**, **OA5**, **OA6** for the fourth to sixth embodiments, $L6T1$ is a thickness of the sixth lenses **L46**, **L56**, **L66** along the optical axes **OA4**, **OA5**, **OA6** for the fourth to sixth embodiments, $L6T2$ is a periphery thickness of the sixth lenses **L46**, **L56**, **L66** for the fourth to sixth embodiments, $Nd6$ is an index of refraction of the sixth lenses **L46**, **L56**, **L66** for the fourth to sixth embodiments, $Nd2$ is an index of refraction of the second lenses **L42**, **L52**, **L62** for the fourth to sixth embodiments, $Nd3$ is an index of refraction of the third lenses **L43**, **L53**, **L63** for the fourth to sixth embodiments, $Nd7$ is an index of refraction of the seventh lenses **L47**, **L57**, **L67** for the fourth to sixth embodiments, $NdLG$ is an index of refraction of the lenses **L45**, **L55**, **L65** closest to the image side for the fourth to sixth embodiments, and $fAS2$ is an effective focal length of the lenses **L47**, **L57**, **L63** second away from the stops **ST4**, **ST5**, **ST6** to the image side for the fourth to sixth embodiments. With the

lens assemblies **4**, **5**, **6** satisfying at least one of the above conditions (2), (4), (7)-(14), the field of view can be effectively increased, the F-number can be effectively decreased, the resolution can be effectively increased, and the aberration can be effectively corrected.

(78) When the condition (7): $65 \text{ mm} \cdot \sin 2^\circ \leq ((f \times \text{FOV} \times \pi) / 360) \times \text{TTL} \leq 70 \text{ mm} \cdot \sin 2^\circ$ is satisfied, the distortion can be corrected effectively.

(79) When the condition (8): $0.4 \leq L6T1/L6T2 \leq 0.8$ is satisfied, the processing cost of lenses can be decreased effectively.

(80) When the condition (9): $1.54 \leq Nd6 \leq 1.68$ is satisfied, the spherical aberration can be corrected effectively.

(81) When the condition (10): $1.54 \leq Nd7 \leq 1.68$ is satisfied, the spherical aberration can be corrected effectively.

(82) When the condition (11): $9 \text{ mm} \leq Nd2 \times f2 \leq 18 \text{ mm}$ is satisfied, the influence of temperature on the offset of the back focal length can be reduced effectively.

(83) When the condition (12): $25 \text{ mm} \leq Nd3 \times f3 \leq 55 \text{ mm}$ is satisfied, the field of curvature can be corrected effectively.

(84) When the condition (13): $20 \text{ mm} \leq NdLG \times fLG \leq 40 \text{ mm}$ is satisfied, the field of curvature can be corrected effectively.

(85) When the condition (14): $13 \text{ mm} \leq fAS2 \leq 40 \text{ mm}$ is satisfied, the distortion can be corrected effectively.

(86) A detailed description of a lens assembly in accordance with a fourth embodiment of the invention is as follows. Referring to FIG. 7, the lens assembly **4** includes a first lens **L41**, a sixth lens **L46**, a second lens **L42**, a stop **ST4**, a third lens **L43**, a seventh lens **L47**, a fourth lens **L44**, a fifth lens **L45**, an optical filter **OF4**, and a cover glass **CG4**, all of which are arranged in order from an object side to an image side along an optical axis **OA4**. In operation, the light from the object side is imaged on an image plane **IMA4**.

(87) According to subject matter previously described herein, wherein: the object side surface **S43** of the sixth lens **L46** is a concave surface and the image side surface **S44** of the sixth lens **L46** is a convex surface; both of the object side surface **S48** and image side surface **S49** of the third lens **L43** are spherical surfaces; the seventh lens **L47** is with positive refractive power, wherein the object side surface **S410** is a convex surface and the image side surface **S411** is a concave surface; the fourth lens **L44** is a biconvex lens, wherein the object side surface **S412** is a convex surface and both of the object side surface **S412** and image side surface **S413** are spherical surfaces; both of the object side surface **S416** and image side surface **S417** of the optical filter **OF4** are plane surfaces, and both of the object side surface **S418** and image side surface **S419** of the cover glass **CG4** are plane surfaces.

(88) With the above design of the lenses, stop **ST4**, and at least one of the conditions (2), (4), (7)-(14) satisfied, the lens assembly **4** can have an effective increased field of view, an effective decreased F-number, an effective increased resolution, and an effective corrected aberration.

(89) Table 10 shows the optical specification of the lens assembly **4** in FIG. 7.

(90) TABLE-US-00010 TABLE 10 Effective Focal Length = 2.53 mm F-number = 1.30 Total Lens Length = 19.08 mm Field of View = 160.40 degrees Effective Radius of Thick- Focal Surface Curvature n Length (mm) (mm) Nd Vd (mm) Remark

S41	10.62	0.79	2.1042	17.02	
-4.540 L41	S42	3.18	2.67	S43	-3.25
1.01	1.6397	23.53	-10.691	L46	S44
-7.12	0.06	S45	19.00		
1.19	2.1042	17.02	7.077	L42	S46
-11.97	0.45	S47	∞	1.30	ST4
S48	-7.22	1.49	1.8061	33.27	
17.194	L43	S49	-5.14	0.08	S410
7.16	2.76	1.5445	56	16.511	L47
S411	32.23	1.08	S412	32.93	
1.51	1.717	47.98	16.829	L44	S413
-18.19	0.07	S414	14.32	1.20	2.1042
17.02	17.916	L45	S415		
55.88	1.45	S416	∞	0.3	1.5168
64.1673	OF4	S417	∞	0.57	S418
∞	0.5	1.5168	64.1673	CG4	S419
∞	0.6				

(91) The aspheric surface sag z of each aspheric lens in table 10 can be calculated by the following formula:

$$z = ch.\sup.2 / \{1 + [1 -$$

$$(k+1)c.\sup.2h.\sup.2].\sup.1/2\}+Ah.\sup.4+Bh.\sup.6+Ch.\sup.8+Dh.\sup.10+Eh.\sup.12$$

where c is curvature, h is the vertical distance from the lens surface to the optical axis, k is conic constant and A , B , C , D , and E are aspheric coefficients.

(92) In the fourth embodiment, the conic constant k and the aspheric coefficients A, B, C, D, E of each aspheric lens are shown in Table 11.

(93) TABLE-US-00011 TABLE 11 Surface Number k A B C D E S43 0 1.33E-02 -4.17E-04
3.49E-05 -1.27E-07 1.62E-07 S44 0 1.04E-02 -4.20E-04 4.23E-05 -1.05E-06 -8.05E-08
S410 0 8.27E-04 5.43E-05 -2.73E-06 6.56E-08 0 S411 0 2.40E-03 8.53E-05 6.93E-07
1.27E-08 0

(94) Table 12 shows the parameters and condition values for conditions (2), (4), (7)-(14) in accordance with the fourth embodiment of the invention. It can be seen from Table 12 that the lens assembly **4** of the fourth embodiment satisfies the conditions (2), (4), (7)-(14).

(95) TABLE-US-00012 TABLE 12 L1D 5.66 mm L6T1 1.01 mm L6T2 1.72 mm NdLG 2.1042 fLG 17.916 mm $((f \times \text{FOV} \times \pi) / 67.47360) \times \text{TTL mm.sup.2} \times \text{degrees}$ L1D/R12 1.78 L6T1/L6T2 0.59 Nd6 1.6397 Nd7 1.5445 Nd2 $\times f2$ 14.89 mm Nd3 $\times f3$ 31.05 mm NdLG $\times fLG$ 37.70 mm fAS2 16.511 mm fAS 5.093 mm ET5 2.11 mm T5 2.76 mm ΦST 4.03 mm ET5/T5 0.76 $\Phi\text{ST/TTL}$ 0.21

(96) In addition, the lens assembly **4** of the fourth embodiment can meet the requirements of optical performance as seen in FIGS. **8A-8D**. It can be seen from FIG. **8A** that the longitudinal aberration in the lens assembly **4** of the fourth embodiment ranges from -0.015 mm to 0.015 mm. It can be seen from FIG. **8B** that the field curvature of tangential direction and sagittal direction in the lens assembly **4** of the fourth embodiment ranges from -0.04 mm to 0.03 mm. It can be seen from FIG. **8C** that the distortion in the lens assembly **4** of the fourth embodiment ranges from -10% to 0% . It can be seen from FIG. **8D** that the lateral color in the lens assembly **4** of the fourth embodiment ranges from -2.1 μm to 2.1 μm . It is obvious that the longitudinal aberration, the field curvature, the distortion, and the lateral color of the lens assembly **4** of the fourth embodiment can be corrected effectively. Therefore, the lens assembly **4** of the fourth embodiment is capable of good optical performance.

(97) Referring to FIG. 9, the lens assembly 5 includes a first lens L51, a sixth lens L56, a second lens L52, a stop ST5, a third lens L53, a seventh lens L57, a fourth lens L54, a fifth lens L55, an optical filter OF5, and a cover glass CG5, all of which are arranged in order from an object side to an image side along an optical axis OA5. In operation, the light from the object side is imaged on an image plane IMA5.

(98) According to subject matter previously described herein, wherein: the object side surface **S53** of the sixth lens **L56** is a concave surface and the image side surface **S54** of the sixth lens **L56** is a convex surface; both of the object side surface **S58** and image side surface **S59** of the third lens **L53** are spherical surfaces; the seventh lens **L57** is with positive refractive power, wherein the object side surface **S510** is a convex surface and the image side surface **S511** is a concave surface; the fourth lens **L54** is a meniscus lens, wherein the object side surface **S512** is a concave surface and both of the object side surface **S512** and image side surface **S513** are aspheric surfaces; both of the object side surface **S516** and image side surface **S517** of the optical filter **OF5** are plane surfaces, and both of the object side surface **S518** and image side surface **S519** of the cover glass **CG5** are plane surfaces.

(99) With the above design of the lenses, stop ST5, and at least one of the conditions (2), (4), (7)-(14) satisfied, the lens assembly 5 can have an effective increased field of view, an effective decreased F-number, an effective increased resolution, and an effective corrected aberration.

(100) Table 13 shows the optical specification of the lens assembly 5 in FIG. 9.

(101) TABLE-US-00013 TABLE 13 Effective Focal Length = 2.55 mm F-number = 1.30 Total Lens Length = 19.00 mm Field of View = 160.40 degrees Effective Radius of Thick- Focal Surface

Curvature nesses Length Number (mm) (mm) Nd Vd (mm) Remark S51 15.22 1.14 1.717 47.98
 -6.798 L51 S52 3.53 2.59 S53 -4.86 1.31 1.6328 23.34 -9.436 L56 S54 -33.49 0.44 SSS 16.12
 1.26 2.1042 17.02 7.818 L52 S56 -16.33 0.08 S57 ∞ 0.55 ST5 S58 -5.51 1.66 1.7859 43.93
 24.776 L53 S59 -4.84 0.08 S510 7.02 1.85 1.5445 56 18.604 L57 S511 21.43 0.73 S512 -26.01
 1.64 1.5445 56 19.698 L54 S513 -7.69 0.56 S514 8.54 1.61 2.1042 17.02 10.032 L55 S515 39.45
 2.48 S516 ∞ 0.3 1.5168 64.1673 OF5 S517 ∞ 0.1 S518 ∞ 0.5 1.5168 64.1673 CG5 S519 ∞
 0.1

(102) The definition of aspheric surface sag z of each aspheric lens in table 13 is the same as that of in Table 10, and is not described here again.

(103) In the fifth embodiment, the conic constant k and the aspheric coefficients A, B, C, D, E of each aspheric lens are shown in Table 14.

(104) TABLE-US-00014 TABLE 14 Surface Number k A B C D E S53 0 9.45E-03 -5.45E-04
 2.33E-05 7.49E-07 -8.08E-08 S54 0 1.05E-02 -3.25E-04 4.03E-05 3.07E-06 -2.82E-07
 S510 0 5.62E-04 3.41E-05 -3.86E-06 7.51E-07 -1.00E-08 S511 0 1.14E-03 4.94E-05
 -5.78E-06 1.23E-06 3.94E-08 S512 0 2.04E-03 1.42E-04 1.18E-05 4.82E-08 -2.35E-08
 S513 0 2.57E-03 1.44E-04 1.29E-05 -3.44E-08 2.65E-08

(105) Table 15 shows the parameters and condition values for conditions (2), (4), (7)-(14) in accordance with the fifth embodiment of the invention. It can be seen from Table 15 that the lens assembly 5 of the fifth embodiment satisfies the conditions (2), (4), (7)-(14).

(106) TABLE-US-00015 TABLE 15 L1D 6.29 mm L6T1 1.31 mm L6T2 2.03 mm NdLG 2.1042
 fLG 10.032 mm $((f \times \text{FOV} \times \pi) / 67.91 \ 360) \times \text{TTL mm} \cdot \text{sup.2} \times \text{degrees}$ L1D/R12 1.78
 L6T1/L6T2 0.65 Nd6 1.6328 Nd7 1.5445 Nd2 $\times$ f2 16.45 mm Nd3 $\times$ f3 44.25 mm NdLG $\times$ fLG
 21.11 mm fAS2 18.604 mm fAS 4.622 mm ET5 1.299 mm T5 1.848 mm Φ ST 3.622 mm ET5/T5
 0.70 Φ ST/TTL 0.19

(107) In addition, the lens assembly 5 of the fifth embodiment can meet the requirements of optical performance as seen in FIGS. 10A-10D. It can be seen from FIG. 10A that the longitudinal aberration in the lens assembly 5 of the fifth embodiment ranges from -0.01 mm to 0.01 mm. It can be seen from FIG. 10B that the field curvature of tangential direction and sagittal direction in the lens assembly 5 of the fifth embodiment ranges from -0.04 mm to 0.01 mm. It can be seen from FIG. 10C that the distortion in the lens assembly 5 of the fifth embodiment ranges from -11% to 0%. It can be seen from FIG. 10D that the lateral color in the lens assembly 5 of the fifth embodiment ranges from -1.8 μm to 1.8 μm . It is obvious that the longitudinal aberration, the field curvature, the distortion, and the lateral color of the lens assembly 5 of the fifth embodiment can be corrected effectively. Therefore, the lens assembly 5 of the fifth embodiment is capable of good optical performance.

(108) Referring to FIG. 11, the lens assembly 6 includes a first lens L61, a sixth lens L66, a stop ST6, a second lens L62, a third lens L63, a seventh lens L67, a fourth lens L64, a fifth lens L65, an optical filter OF6, and a cover glass CG6, all of which are arranged in order from an object side to an image side along an optical axis OA6. In operation, the light from the object side is imaged on an image plane IMA6.

(109) According to subject matter previously described herein, wherein: the object side surface S63 of the sixth lens L66 is a convex surface and the image side surface S64 of the sixth lens L66 is a concave surface; both of the object side surface S68 and image side surface S69 of the third lens L63 are aspheric surfaces; the seventh lens L67 is with negative refractive power, wherein the object side surface S610 is a concave surface and the image side surface S611 is a convex surface; the fourth lens L64 is a meniscus lens, wherein the object side surface S612 is a concave surface and both of the object side surface S612 and image side surface S613 are aspheric surfaces; both of the object side surface S616 and image side surface S617 of the optical filter OF6 are plane surfaces, and both of the object side surface S618 and image side surface S619 of the cover glass CG6 are plane surfaces.

(110) With the above design of the lenses, stop ST6, and at least one of the conditions (2), (4), (7)-(14) satisfied, the lens assembly 6 can have an effective increased field of view, an effective decreased F-number, an effective increased resolution, and an effective corrected aberration.

(111) Table 16 shows the optical specification of the lens assembly 6 in FIG. 11.

(112) TABLE-US-00016 TABLE 16 Effective Focal Length = 2.57 mm F-number = 1.30 Total Lens Length = 18.46 mm Field of View = 160.70 degrees Effective Radius of Thick- Focal Surface Curvature n Length Number (mm) (mm) Nd Vd (mm) Remark S61 13.85 1.32 1.5688 56.04 -8.888 L61 S62 3.54 1.74 S63 19.59 1.00 1.5365 55.98 -7.113 L66 S64 3.10 1.29 S65 ∞ -0.04 ST6 S66 32.00 2.16 2.1042 17.02 4.570 L62 S67 -5.50 0.08 S68 -4.18 1.34 1.5445 56 34.434 L63 S69 -3.79 0.05 S610 -5.99 0.85 1.6613 20.38 -91.508 L67 S611 -7.04 0.07 S612 -17.86 2.56 1.5445 56 10.950 L64 S613 -4.65 0.08 S614 13.00 2.26 2.1042 17.02 17.266 L65 S615 41.00 2.70 S616 ∞ 0.3 1.5168 64.1673 OF6 S617 ∞ 0.1 S618 ∞ 0.5 1.5168 64.1673 CG6 S619 ∞ 0.1

(113) The definition of aspheric surface sag z of each aspheric lens in table 16 is the same as that of in Table 10, and is not described here again.

(114) In the sixth embodiment, the conic constant k and the aspheric coefficients A, B, C, D, E of each aspheric lens are shown in Table 17.

(115) TABLE-US-00017 TABLE 17 Surface Number k A B C D E S63 0 2.64E-03 1.37E-05 -8.42E-05 1.16E-05 -5.33E-07 S64 0 1.59E-02 -1.41E-03 1.66E-03 -5.04E-04 1.10E-04 S68 0 3.48E-03 1.96E-03 -7.44E-04 1.50E-04 -9.23E-06 S69 0 -5.86E-04 -6.06E-04 2.36E-04 -4.53E-05 5.06E-06 S610 0 -3.40E-03 -1.16E-04 1.04E-05 3.97E-07 3.63E-08 S611 0 1.50E-03 9.12E-05 1.03E-06 -1.01E-07 -1.03E-07 S612 0 -1.47E-03 -1.22E-04 -6.51E-06 -9.92E-07 2.23E-08 S613 0 -4.98E-04 1.64E-05 -4.93E-06 -3.12E-08 -1.82E-09

(116) Table 18 shows the parameters and condition values for conditions (2), (4), (7)-(14) in accordance with the sixth embodiment of the invention. It can be seen from Table 18 that the lens assembly 6 of the sixth embodiment satisfies the conditions (2), (4), (7)-(14).

(117) TABLE-US-00018 TABLE 18 L1D 6.26 mm L6T1 1.00 mm L6T2 1.52 mm NdLG 2.1042 fLG 17.266 mm ((f × FOV × π)/ 66.54 360) × TTL mm.sup.2 × degrees L1D/R12 1.77 L6T1/L6T2 0.66 Nd6 1.5365 Nd7 1.6613 Nd2 × f2 9.62 mm Nd3 × f3 53.18 mm NdLG × fLG 36.33 mm fAS2 34.434 mm fAS 3.774 mm ET5 1.048 mm T5 0.846 mm ΦST 3.146 mm ET5/T5 1.24 ΦST/TTL 0.17

(118) In addition, the lens assembly 6 of the sixth embodiment can meet the requirements of optical performance as seen in FIGS. 12A-12D. It can be seen from FIG. 12A that the longitudinal aberration in the lens assembly 6 of the sixth embodiment ranges from -0.02 mm to 0.03 mm. It can be seen from FIG. 12B that the field curvature of tangential direction and sagittal direction in the lens assembly 6 of the sixth embodiment ranges from -0.03 mm to 0.05 mm. It can be seen from FIG. 12C that the distortion in the lens assembly 6 of the sixth embodiment ranges from -12% to 0%. It can be seen from FIG. 12D that the lateral color in the lens assembly 6 of the sixth embodiment ranges from -1.6 μm to 1.6 μm. It is obvious that the longitudinal aberration, the field curvature, the distortion, and the lateral color of the lens assembly 6 of the sixth embodiment can be corrected effectively. Therefore, the lens assembly 6 of the sixth embodiment is capable of good optical performance.

(119) While the invention has been described by way of example and in terms of the preferred embodiment(s), it is to be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

Claims

1. A lens assembly comprising: a first lens which is a meniscus lens with negative refractive power; a second lens which is with positive refractive power and comprises a convex surface facing an image side; a third lens which is a meniscus lens with positive refractive power; a fourth lens which is with positive refractive power; and a fifth lens which is with refractive power and comprises a convex surface facing an object side; wherein the first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from the object side to the image side along an optical axis; wherein the lens assembly satisfies:
 $0.1 < \Phi_{ST}/TTL < 0.5$; wherein Φ_{ST} is an aperture diameter of a stop and TTL is an interval from an object side surface of the first lens to an image plane along the optical axis.
2. The lens assembly as claimed in claim 1, wherein: the first lens comprises a convex surface facing the object side and a concave surface facing the image side; the second lens further comprises a concave surface facing the object side; and the third lens comprises a concave surface facing the object side and a convex surface facing the image side.
3. The lens assembly as claimed in claim 2, wherein: the fourth lens is a meniscus lens and comprises a convex surface facing the object side and a concave surface facing the image side; and the fifth lens is a biconvex lens with positive refractive power and further comprises a convex surface facing the image side.
4. The lens assembly as claimed in claim 1, wherein the stop is disposed between the second lens and the third lens.
5. The lens assembly as claimed in claim 1, wherein the fifth lens is a single lens and an air gap is disposed between the fourth lens and the fifth lens.
6. The lens assembly as claimed in claim 1, wherein the lens assembly satisfies:
 $-3 < f_1/f < -1.5$;
 $0.5 < ET_5/T_5 < 1.5$; wherein f_1 is an effective focal length of the first lens, f is an effective focal length of the lens assembly, ET_5 is a periphery thickness of the fifth lens, and T_5 is a thickness of the fifth lens along the optical axis.
7. The lens assembly as claimed in claim 1, further comprising a sixth lens and a seventh lens, wherein: the sixth lens is a meniscus lens with negative refractive power and disposed between the first lens and the second lens; the seventh lens is a meniscus lens and disposed between the third lens and the fourth lens; and the fifth lens is a meniscus lens with positive refractive power and further comprises a concave surface facing the image side.
8. The lens assembly as claimed in claim 7, wherein: the first lens comprises a convex surface facing the object side and a concave surface facing the image side; the second lens further comprises another convex surface facing the object side; the third lens comprises a concave surface facing the object side and a convex surface facing the image side; and the fourth lens comprises a convex surface facing the image side.
9. The lens assembly as claimed in claim 8, wherein: the sixth lens comprises a concave surface facing the object side and a convex surface facing the image side; and the seventh lens is with positive refractive power and comprises a convex surface facing the object side and a concave surface facing the image side.
10. The lens assembly as claimed in claim 9, wherein the fourth lens further comprises another convex surface facing the object side.
11. The lens assembly as claimed in claim 9, wherein the fourth lens further comprises a concave surface facing the object side.
12. The lens assembly as claimed in claim 8, wherein: the sixth lens comprises a convex surface facing the object side and a concave surface facing the image side; and the seventh lens is with negative refractive power and comprises a concave surface facing the object side and a convex surface facing the image side.
13. The lens assembly as claimed in claim 12, wherein the fourth lens further comprises a concave

surface facing the object side.

14. The lens assembly as claimed in claim 7, wherein the lens assembly satisfies at least one of following conditions:

$$65 \text{ mm} \cdot \sup.2 \times \text{degrees} \leq ((f \times FOV \times \pi) / 360) \times TTL \leq 70 \text{ mm} \cdot \sup.2 \times \text{degrees};$$

$$L1D/R12 \leq 1.8;$$

$$0.4 \leq L6T1/L6T2 \leq 0.8;$$

$$1.54 \leq Nd6 \leq 1.68;$$

$$1.54 \leq Nd7 \leq 1.68;$$

$9 \text{ mm} \leq Nd2 \times f2 \leq 18 \text{ mm}$; wherein f is an effective focal length of the lens assembly, FOV is a field of view of the lens assembly, TTL is an interval from an object side surface of the first lens to an image plane along the optical axis, $L1D$ is an effective optical diameter of an image side surface of the first lens, $R12$ is a radius of curvature of the image side surface of the first lens, $L6T1$ is a thickness of the sixth lens along the optical axis, $L6T2$ is a periphery thickness of the sixth lens, $Nd6$ is an index of refraction of the sixth lens, $Nd7$ is an index of refraction of the seventh lens, $Nd2$ is an index of refraction of the second lens, and $f2$ is an effective focal length of the second lens.

15. The lens assembly as claimed in claim 7, wherein the stop is disposed between the sixth lens and the third lens and the lens assembly satisfies at least one of following conditions:

$$25 \text{ mm} \leq Nd3 \times f3 \leq 55 \text{ mm};$$

$$20 \text{ mm} \leq NdLG \times fLG \leq 40 \text{ mm};$$

$$13 \text{ mm} \leq fAS2 \leq 40 \text{ mm};$$

$3.5 \text{ mm} \leq fAS \leq 5.5 \text{ mm}$; wherein $Nd3$ is an index of refraction of the third lens, $f3$ is an effective focal length of the third lens, $NdLG$ is an index of refraction of the lens closest to the image side, fLG is an effective focal length of the lens closest to the image side, $fAS2$ is an effective focal length of the lens second away from the stop to the image side, and fAS is an effective focal length of a combination of all lenses between the stop and the image side.

16. A lens assembly comprising: a first lens which is a meniscus lens with negative refractive power; a second lens which is with positive refractive power and comprises a convex surface facing an image side; a third lens which is a meniscus lens with positive refractive power; a fourth lens which is with positive refractive power; and a fifth lens which is with refractive power; wherein the first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from an object side to the image side along an optical axis; wherein the lens assembly satisfies at least one of following conditions:

$$0.5 < ET5/T5 < 1.5;$$

$65 \text{ mm} \cdot \sup.2 \times \text{degrees} \leq ((f \times FOV \times \pi) / 360) \times TTL \leq 70 \text{ mm} \cdot \sup.2 \times \text{degrees}$; Wherein $ET5$ is a periphery thickness of the fifth lens, $T5$ is a thickness of the fifth lens along the optical axis, f is an effective focal length of the lens assembly, FOV is a field of view of the lens assembly, and TTL is an interval from an object side surface of the first lens to an image plane along the optical axis.

17. The lens assembly as claimed in claim 16, further comprising a stop disposed between the second lens and the third lens, wherein: the first lens comprises a convex surface facing the object side and a concave surface facing the image side; the second lens is a meniscus lens and further comprises a concave surface facing the object side; the third lens comprises a concave surface facing the object side and a convex surface facing the image side; the fourth lens is a biconvex lens and comprises a convex surface facing the object side and another convex surface facing the image side, and the fifth lens is a meniscus lens and comprises a concave surface facing the object side and a convex surface facing the image side.

18. The lens assembly as claimed in claim 17, wherein the lens assembly satisfies at least one of following conditions:

$$0.1 < \Phi_{ST}/TTL < 0.5;$$

$$-3 < f1/f < -1.5;$$

$0.5 < ET5/T5 < 1.5$;

$0.3 < \Phi2/\Phi3 < 0.8$; wherein Φ_{ST} is an aperture diameter of the stop, TTL is an interval from an object side surface of the first lens to an image plane along the optical axis, f_1 is an effective focal length of the first lens, f is an effective focal length of the lens assembly, $ET5$ is a periphery thickness of the fifth lens, $T5$ is a thickness of the fifth lens along the optical axis, $\Phi2$ is a maximum effective optical diameter of the second lens, and $\Phi3$ is a maximum effective optical diameter of the third lens.

19. The lens assembly as claimed in claim 16, wherein: the fifth lens is with negative refractive power; and the lens assembly satisfies:

$0.3 < \Phi2/\Phi3 < 0.8$; wherein $\Phi2$ is a maximum effective optical diameter of the second lens, and $\Phi3$ is a maximum effective optical diameter of the third lens.
